

EBS – Elastic Block Storage

Class no: 14

EBS:

Amazon EBS (Elastic Block Store) is block-level storage for EC2 instances. Root volumes must be created in the same Availability Zone as the EC2 instance, with typical defaults of 8 GB for Linux and 30 GB for Windows. EBS offers SSD and HDD families, each optimized for different workloads, with gp3 being the latest general-purpose SSD type.

Root Volume Sizes

- **Linux EC2 instances:** Default root volume size is **8 GB**.
- **Windows EC2 instances:** Default root volume size is **30 GB**.
- You can increase these sizes during instance launch or later by resizing the volume.

Important: The root EBS volume **must be in the same Availability Zone** as the EC2 instance to attach successfully.

EBS Volume Types

EBS volumes are categorized into **SSD-based** (optimized for performance) and **HDD-based** (optimized for throughput and cost).

Type	Family Members	Purpose / Use Case
General Purpose SSD (gp2, gp3)	gp2, gp3	Balanced price/performance. Boot volumes, dev/test, medium DBs, interactive apps. gp3 allows independent scaling of IOPS & throughput.
Provisioned IOPS SSD (io1, io2)	io1, io2	High-performance, mission-critical DBs needing consistent low latency and high IOPS.
Throughput Optimized HDD (st1)	st1	Big data, data warehouses, log processing. Optimized for sequential reads/writes.
Cold HDD (sc1)	sc1	Lowest-cost option. Infrequently accessed data, archival workloads.

Key Notes on gp3 (Latest SSD Family)

- **Baseline performance:** 3,000 IOPS regardless of size.
- **Scalable:** You can provision up to **16,000 IOPS** and **1,000 MB/s throughput** independently of storage size.
- **Cost efficiency:** ~20% lower price per GB compared to gp2.
- **Best for:** General workloads, boot volumes, and applications needing predictable performance.

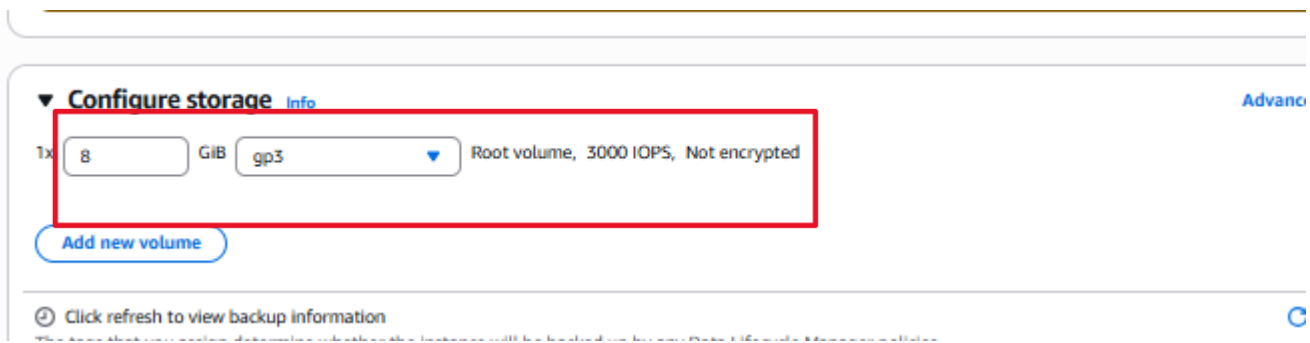
{ Backup of EBS (volume) is called snapshot.

Similarly, Backup of Ec2 is called AMI (Amazon Machine Image).}

Hands on: creating a Linux Ec2 instance and mount new extra volume:

Step no:1 Create a Linux Ec2 instance with the default root volume of 8gb. Launch it and connect it.

Note: AWS doesn't allow us to choose HDD as root volume, by default it allows SSD.



Step no:2 Now connect with the EC2 instance. Check the available block storage using: **lsblk**

Followed by , **df -h** -> to check the disk free space.

```
[ec2-user@ip-172-31-33-210 ~]$ sudo su
[root@ip-172-31-33-210 ec2-user]# pwd
/home/ec2-user
[root@ip-172-31-33-210 ec2-user]# lsblk
NAME                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
nvme0n1              259:0    0   8G  0 disk
├─nvme0n1p1          259:1    0   8G  0 part /
├─nvme0n1p127        259:2    0  1M  0 part
└─nvme0n1p128        259:3    0 10M  0 part /boot/efi
[root@ip-172-31-33-210 ec2-user]# df h
df: h: No such file or directory
[root@ip-172-31-33-210 ec2-user]# df -h
Filesystem            Size  Used Avail Use% Mounted on
devtmpfs               4.0M    0  4.0M   0% /dev
tmpfs                  459M    0  459M   0% /dev/shm
tmpfs                  184M  432K  183M   1% /run
/dev/nvme0n1p1         8.0G  1.6G  6.4G  20% /
tmpfs                  459M    0  459M   0% /tmp
/dev/nvme0n1p128       10M  1.3M  8.7M  13% /boot/efi
tmpfs                   92M    0   92M   0% /run/user/1000
```

We can see the default root volume and available free space.

(**Note:** We cannot reduce the size of EBS volume. While creation we have to be careful. But we can expand the size.)

Step no:3 Now we are going to add a 20gb additional storage. First create a EBS volume. Go to the instance page, scroll down → EBS → Create volume.

Follow the configurations:

- 20 gb
- 3000 IOPS
- 125 throughputs
- Availability zone must be same as of our EC2 instance
- Don't create a volume from the snapshot
- Create volume

EC2

Dashboard
AWS Global View
Events
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AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager

Volumes (1) Info

Saved filter sets

Choose filter set

Q Search

	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Source volume ID	Created	Availability Zone	Volume state
		vol-0903b323112a5dce4	gp3	8 GiB	3000	125	snap-04e7ac1...	-	2026/03/30 18:47 GMT+5:...	eun1-az2 (eu-north-...	In-use

Recycle Bin

Actions

Create volume

Fault tolerance for all volumes in this Region

Snapshot summary

Recently backed up volumes / Total # volumes

0 / 1

Last updated on Mon, Mar 30, 2026, 07:05:21 PM (GMT+05:30)

Data Lifecycle Manager default policy for EBS Snapshots status

No default policy set up | Create policy

Create volume

Create an Amazon EBS volume to attach to any EC2 Instance in the same Availability Zone.

Volume settings

Volume type

General Purpose SSD (gp3)

Size (GiB)

20

Min: 1 GiB, Max: 65536 GiB.

IOPS

3000

Min: 3000 IOPS, Max: 80000 IOPS.

Throughput (MiB/s)

125

Min: 125 MiB, Max: 2000 MiB. Baseline: 125 MiB/s.

Availability Zone

eun1-az2 (eu-north-1b)

Snapshot ID - optional

Don't create volume from a snapshot

Encryption

Use Amazon EBS encryption as an encryption solution for your EBS resources associated with your EC2 instances.

☐ Encrypt this volume

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

Add new tag

You can add up to 50 tags.

Snapshot summary

Click refresh to view backup information

The volume type that you select and the tags that you assign determine whether the volume will be backed up by any Data Lifecycle Manager policies.

Cancel

Create volume

Step no:4 Once created, the volume shows **Available** status.

Volumes (2) Info											
Saved filter sets Choose filter set Q Search											
	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Source volume ID	Created	Availability Zone	Volume state	
	vol-0903b323112a5dce4	gp3	8 GiB	3000	125	snap-04e7ac1...	-	2026/03/30 18:47 GMT+5:...	eun1-az2 (eu-north-...	In-use	
	vol-0e67333a18e71b278	gp3	20 GiB	3000	125	-	-	2026/03/30 19:14 GMT+5:...	eun1-az2 (eu-north-...	Available	

As it is not attached with any EC2. Now click the volume → action → attach volume

Successfully created volume vol-0e67333a18e71b278.

Volumes (1/2) Info

Save filter sets Choose filter set Search

	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Source volume ID	Created	Actions
☐		vol-0903b323112a5dce4	gp3	8 GiB	3000	125	snap-04e7ac1...	-	20...	
☒		vol-0e67333a18e71b278	gp3	20 GiB	3000	125	-	-	20...	Attach volume

Modify volume
Create snapshot
Create snapshot lifecycle policy
Delete volume
Attach volume
Detach volume
Force detach volume
Manage auto-enabled I/O
Copy volume - new

→ choose our instance (it should be in running state) → path as `/dev/sdb` → Attach volume.

Attach volume Info

Attach a volume to an instance to use it as you would a regular physical hard disk drive.

Basic details

Volume ID
vol-0e67333a18e71b278

Availability Zone
eu-n1-az2 (eu-north-1b)

Instance Info
i-073d414f9b5da440f (My Linux Machine) (running)

Device name Info
/dev/sdb

Recommended device names for Linux: /dev/sda for root volume, /dev/sd[f-p] for data volumes.

Newer Linux kernels may rename your devices to /dev/xvdf through /dev/xvdp internally, even when the device name entered here (and shown in the details) is /dev/sdf through /dev/sdp.

Cancel [Attach volume](#)

→ Now the status will be **in use**.

2026/03/30 18:47 GMT+5:...	eu-n1-az2 (eu-north-...	✓ In-use	No alarms	+	i-073d414f9b5da440f (My ...
2026/03/30 19:14 GMT+5:...	eu-n1-az2 (eu-north-...	✓ In-use	No alarms	+	i-073d414f9b5da440f (My ...

Step no:4 Even though we have attached the additional volume to the EC2 it will not be available for usage, until we format it.

```
NAME                MAJ:MIN RM  SIZE RO  TYPE MOUNTPOINTS
nvme0n1             259:0    0   8G  0  disk
├─nvme0n1p1         259:1    0   8G  0  part /
├─nvme0n1p127       259:2    0    1M  0  part
└─nvme0n1p128       259:3    0   10M  0  part /boot/efi
nvme1n1             259:4    0  20G  0  disk
[ec2-user@ip-172-31-33-210 ~]$ df -h
Filesystem          Size  Used Avail Use% Mounted on
devtmpfs            4.0M    0  4.0M   0% /dev
tmpfs               459M    0  459M   0% /dev/shm
tmpfs               184M  444K  183M   1% /run
/dev/nvme0n1p1      8.0G  1.6G  6.4G  20% /
tmpfs               459M    0  459M   0% /tmp
/dev/nvme0n1p128    10M  1.3M  8.7M  13% /boot/efi
tmpfs                92M    0   92M   0% /run/user/1000
```

We can see this is available as a storage, but not under the disk usage.

Step no:5 To make it available in usage, we need to format it.

First check whether it is mounted or not: **file -s /dev/nvme1n1**

```
[root@ip-172-31-33-210 ec2-user]# file -s /dev/nvme1n1
/dev/nvme1n1: data
```

Step no:6 We can see the data is not formatted. Now we have to format it in Linux Filesystem called XFS. Using:

mkfs -t xfs /dev/nvme1n1

(There is also Ext4, which is also a linux file system formatting.)

```
[root@ip-172-31-33-210 ec2-user]# mkfs -t xfs /dev/nvme1n1
meta-data=/dev/nvme1n1            isize=512    agcount=16, agsize=327680 blks
                                     =               sectsz=512    attr=2, projid32bit=1
                                     =               crc=1        finobt=1, sparse=1, rmapbt=0
                                     =               reflink=1   bigtime=1 inobtcount=1 nrext64=0
                                     =               exchange=0
data      =                       bsize=4096   blocks=5242880, imaxpct=25
                                     =               sunit=1      swidth=1 blks
naming    =version 2              bsize=4096   ascii-ci=0, ftype=1, parent=0
log        =internal log          bsize=4096   blocks=16384, version=2
                                     =               sectsz=512   sunit=1 blks, lazy-count=1
realtime   =none                  extsz=4096   blocks=0, rtextents=0
```

Step no:7 As the file system has been formatted, now we have to mount it to our directory.

First create a directory under mount where the additional volumes will be added: **mkdir /mnt/myvolume**

Now mount the volume to our directory: **mount /dev/nvme1n1 /mnt/myvolume**
(mount source destination)

```
[root@ip-172-31-33-210 ec2-user]# mkdir /mnt/myvolume
[root@ip-172-31-33-210 ec2-user]# mount /dev/nvme1n1 /mnt/myvolume
```

Step no:8 we can check the process is successful by using: **df -h**

Where now we can see the additional storage under usage.

```
[root@ip-172-31-33-210 ec2-user]# df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M   0    4.0M   0% /dev
tmpfs           459M   0    459M   0% /dev/shm
tmpfs           184M  440K   183M   1% /run
/dev/nvme0n1p1   8.0G  1.6G   6.4G  20% /
tmpfs           459M   0    459M   0% /tmp
/dev/nvme0n1p128 10M   1.3M   8.7M  13% /boot/efi
tmpfs           92M   0     92M   0% /run/user/1000
/dev/nvme1n1     20G  176M   20G   1% /mnt/myvolume
```

Step no:9 If we stop and restart the instance , the added volume will be detached from our respective instance, to avoid that we must make an entry in the **/etc/fstab**.

Command: **nano /etc/fstab**

```
[root@ip-172-31-33-210 ec2-user]# nano /etc/fstab
```

Now add **/dev/nvme1n1 /data xfs default** → under the file.

Press: Cntrl+c and Cntrl+Y → to save and exit.

```
GNU nano 8.3 /etc/fstab
#
UUID=a7e0b2a5-da35-4366-bb65-556fc198bb3f / xfs defaults,noatime 1 1
UUID=8CC6-72F8 /boot/efi vfat defaults,noatime,uid=0,gid=0,umask=0077,shortname=winnt,x-systemd.automount 0 2
/dev/nvme1n1 /data xfs default
```